MILITARY SPECIFICATION

TEST SET, RADAR AN/UPM-116A

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be a portable test set capable of giving a "go-no-go" check on the operation of Radar Set AN/APQ-83 or Radar Set AN/APQ-94 on the test bench or during preflight check of the aircraft.

1.2 Classification - The Test Set, Radar AN/UPM-116A shall be of one type and shall consist of the following items:

Item	Type Designation	Appl. Para.
Test Set, Radar	TS-1414A/UPM-116	3.3.1
Cable Assembly, Special Purpose	W-1 of AN/UPM-116	3.3.2
Cable Assembly, Radio Frequency	W-2 of AN/UPM-116	3.3.3
Cable Assembly, Radio Frequency	W-3 of AN/UPM-116	3.3.3
Cover, Test Set	CW-561/UPM-116	3.2.4.2

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-21642 Radar Set AN/APQ-83, Military Specification for

MIL-R-23373 Radar Set AN/APQ-94, Military Specification for

MIL-E-4682 Electron Tubes and Transistors, Choice and Application of

FSC 6625

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MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-D-18300	Design Data Requirements for Contracts Covering Airborne Avionic Equipment
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

PUBLICATIONS

SAR-300	Material Changes and Material Bulletins, Preparation of

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

2.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

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3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment, except as modified herein.

3.2.2.1 Vibration - In accordance with Specification MIL-T-21200, paragraph 3.2.14.4 with these exceptions:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 33	0.02 inch

A test at 0.02 inch double amplitude with the rate of frequency varying between 33 and 55 cycles per second is to be run for exploratory determination of mechanical transmissibility. This test is to be non-destructive.

3.2.2.2 Sand, Dust and Fungus - Sand, dust and fungus requirements of Specification MIL-T-21200 shall not be applicable to this equipment.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment:

Temperature	Room Ambient ($30^{o} \pm 10^{o}$C)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% Relative Humidity
Input Power Voltage	27 ± 2 VDC

3.2.4 Form Factor - The Test Set, Radar AN/UPM-116 shall not exceed 14 inches wide by 10-1/4 inches high by 10-1/2 inches deep, and shall not exceed 25 pounds excluding cables.

3.2.4.1 Case - The combination case shall conform to Specification MIL-T-21200.

3.2.4.2 Cover, Test Set CW-561/UPM-116 - The cover shall be detachable and shall contain provision for stowing an instruction book and the cables per Specification MIL-T-21200.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 24 hours continuously or intermittently, without the necessity for readjustment of any controls.

3.2.6 Operating Life -

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

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3.2.7 Under Voltage Protection - The equipment shall not be damaged by operation at 21 volts and shall automatically resume normal operation when the voltage returns within limits.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-T-21200.

3.2.10 Maintenance Provision and Field Testing - Provision for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.11 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.12 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200.

3.3 Detail Requirements -

3.3.1 Test Set, Radar TS-1414A/UPM-116 - The radar test set shall meet the following requirements:

3.3.1.1 Function - The test set shall perform a preflight check on Radar Set AN/APQ-83 (as specified in Specification MIL-R-21642) or Radar Set AN/APQ-94 (as specified in Specification MIL-R-23373) while installed in the aircraft or on the associated test bench.

3.3.1.2 Finish - The case shall be finished with gray enamel conforming to MIL-T-21200. The panel shall be finished with black synthetic lusterless enamel in accordance with TT-E-527.

3.3.1.3 Contents of Unit - The test set shall contain the necessary circuits to perform the following functions:

3.3.1.3.1 Null Meter Voltage Measurements - The test set shall measure the following power supply voltages on a "go-no-go" basis to an accuracy of ± 1%.

(1) +500 VDC
(2) +300 VDC
(3) +150 VDC
(4) -250 VDC
(5) +20 VDC
(6) -20 VDC

In addition, the test set shall measure (7) range and (8) range rate voltages corresponding to 17,000 yards and 360 knots (closing) on a "go-no-go" basis to an accuracy of ± 1%.

3.3.1.3.2 Current Measurement - The test set shall measure the following currents to an accuracy as noted:

(1)	Magnetron Current	± 5% (go-no-go)
(2)	#1 IF Crystal Current	± 5% (go-no-go)
(3)	#2 IF Crystal Current	± 5% (go-no-go)
(4)	#1 AFC Crystal Current	± 5% (go-no-go)
(5)	#2 AFC Crystal Current	± 5% (go-no-go)

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3.3.1.3.3 Velocity Test - The test set shall provide means to test the tracking and search rate of the range tracker of the Radar Set as noted below:

(1) A DC voltage (controlled from the front panel) representing a closing target velocity of 300 to 500 knots is switched to the range tracker. The range rate voltage as measured by the radar test set will ascertain that the range computer is capable of tracking a 300 to 500 knot target.

(2) With the radar set in the range mode of operation, the test set is used to measure the range rate voltage to ascertain the sweep rate of the range track gate.

3.3.1.3.4 Range Volts Measurement - The test set shall be capable of measuring the range volts output of the Radar Set.

3.3.1.3.5 Receiver Tests - The test set shall be capable of:

(1) Measuring the receiver's ability to detect modulation. (The amount of modulation to be controlled from the front panel.)

(2) Measuring the AGC operation of the Radar Set.

3.3.1.3.6 Antenna Position Test - The test set shall be capable of positioning the antenna in four quadrants and indicating antenna position on the multimeter.

3.3.1.3.6.1 The test set shall be capable of energizing the scan failure relay in the radar set.

3.3.1.3.7 Transmitter Disable Function - The test set shall provide a means of disabling the magnetron in the Radar Set. At the same time a 30 mc pulse shall be applied to the Radar Set to trigger the 1000 yard marker generator.

3.3.1.3.7.1 30 MC Pulsed Oscillator - The test set shall provide a 30 MC pulsed oscillator with the following characteristics:

Output Pulse:

Frequency	30 ± 1.0 MC
Amplitude	0.635 volt peak to peak minimum into a 50 ohm ± 10% load.
Width	0.6 to 0.9 usec.
PRF	Same as input trigger pulse

Input Trigger Pulse: The 30 MC pulsed oscillator shall be triggered by the radar system trigger which is defined as follows:

Amplitude	20 volts positive (minimum, unloaded)
Width	Less than 10 usec. at half amplitude
Rise Time	Less than 0.5 usec.
PRF	1200 cps ± 10%

3.3.1.3.8 Manual Tune Function - The test set shall provide a means of manually tuning the klystron in the Radar Set.

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3.3.1.3.9 Test Targets Function - The test set shall provide a means of energizing the 1000 yard marker generator in the Radar Set.

3.3.1.4 Front Panel Layout - All operational controls and indicator shall be located on the front panel. Prior to fabrication of the first equipment, the contractor shall submit a drawing of the proposed panel layout, including identifying title and calibration marks, to the procuring activity for approval.

3.3.1.4.1 Controls and Indicators - The following controls and indicators shall be provided on the front panel of the test set:

(1) Multimeter - A 2-1/2 inch sealed and ruggedized panel meter shall be provided. It shall have a 0-100 microamp movement with the meter scale calibrated as follows with an accuracy at full scale within ± 2%.

(a) 0 to 40,000 yards

(b) 0 to 500 knots

(c) 0 to 10 volts

(d) 0 to 50 degrees

(e) Magnetron current with a parameter at 18.3 (min.) and 28.6 ma (max.)

(2) Multimeter Function Switch - The multimeter function switch shall switch the multimeter circuitry to provide full scale deflection ± 1/2 scale division (± 2.5%) for inputs indicated for each of the following labeled switch positions:

(a)	"POS RANGE RATE (CLOSING)"	+2.78 VDC
(b)	"NEG RANGE RATE (CLOSING)"	- 2.78 VDC
(c)	"AGC VOLTS"	- 10 VDC
(d)	"MAG CUR"	+28.6 MA
(e)	"RANGE"	+200 VDC
(f)	"+SIN EL"	+68.3 VDC
(g)	"- SIN EL"	- 68.3 VDC
(h)	"+SIN AZ"	+68.3 VDC
(i)	"- SIN AZ"	- 68.3 VDC
(j)	"OFF"	

(3) Null Meter - A 2-1/2 inch sealed and ruggedized panel meter shall be provided. It shall have a 100-0-100 microamp nonlinear movement. The scale shall be marked with three bands: ± 5%, ± 2%, and ± 1%. These bands shall indicate the percent deviation from normal of the signals being measured. The operation of the meter shall be such that the input signals are compared to reference voltages with an initial accuracy of ± 0.1%.

(4) Null Meter Function Switch - The null meter function switch shall switch the null meter circuitry (0.5 MA load) and shall have the following labeled positions which, unless otherwise indicated, are the input signal voltages:

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(a) "+500" Scaling resistors internal to radar set

(b) "+300"

(c) "+150"

(d) "- 250"

(e) "+20"

(f) "- 20"

(g) "RANGE RATE" -2.0 VDC input signal corresponding to 360 knots closing.

(h) "RANGE" +85 VDC input signal corresponding to 17,000 yards

(5) Crystal Current Meters - Four one-inch sealed and ruggedized panel meters shall be provided for continuously monitoring the four crystal currents. They shall have 0-100 microamp movements. The scales shall be marked with parameters indicating 0.7 ma (min) and 1.2 ma (max) and shall be calibrated with an accuracy of ± 2% of full scale deflection at the 0.9 ma scale division.

(6) Antenna Position Switch - The antenna position switch shall switch the modulation input voltages and provide a means of energizing the scan failure relay in the Radar Set and shall have the following labeled positions:

(a) "SCAN FAIL"

(b) "OFF"

(c) "UP"

(d) "RIGHT"

(e) "DOWN"

(f) "LEFT"

(7) Range Targets Switch - A switch labeled "RANGE TARGETS" "ON-OFF" shall be provided to connect +28 VDC to the test targets relay in the Radar Set.

(8) Meter Sensitivity Switch - A switch labeled "METER SENSITIVITY" "MAX-MIN" shall be provided to protect the Null Meter against overloads. This is a spring loaded toggle switch which is normally in the Min. sensitivity position. Operation of this switch to the "MAX" position removes a shunt resistor from the Null Meter circuit.

(9) Antenna Test Switch - A switch labeled "TEST" "AGC-OFF-TRACK" shall be provided to connect +28 VDC to the Angle Track Test Relay and the AGC Test Relay in the Radar Set.

(10) Transmitter Disable Switch - A switch labeled "XMTR" "OFF-NORMAL" shall be provided to connect +28 VDC to the transmitter disable relay in the Radar Set and at the same time to turn on the 30 mc pulsed oscillator in the test set.

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(11) Range Rate Test Switch - A switch labeled "RANGE RATE TEST" shall be provided to operate in conjunction with the multimeter and multimeter function switch in the neg. range rate (closing) position and shall have the following labeled positions:

(a) "TARGET RATE" Sets multimeter scale factor for -2.78 VDC full scale representing 500 knots closing.

(b) "STROBE RATE" Sets multimeter scale factor for +40 VDC full scale for range strobe rate check.

(c) "VELOCITY TEST" Functions same as (a) but also connects voltage to Radar Set range rate test circuit.

(12) Range Rate Control - A control labeled "RANGE RATE" "INCREASE-DECREASE" shall be provided to adjust the level of the DC voltage connected to the range rate section of the range computer. The range of the control shall be such that target rates between 300 and 500 knots closing can be simulated. The voltage output of this control shall be between a minimum of +0.73 ± 0.1 VDC and a maximum of +1.40 ± 0.07 VDC when connected to a 500 ohm ± 0.1% resistive load.

(13) Tune Control - A control labeled "TUNE" "AUTO-MANUAL" shall be provided to allow manual adjustment of the klystron repeller voltage and shall have a range of from -78 ± 5.0 VDC to -158 ± 10 VDC. This control shall be provided with a switch which shall connect +28 VDC to the Radar Set auto-manual tune relay except when the control is in its most CCW position.

(14) Modulation Control - A control labeled "MODULATION" shall be provided to adjust the level of the 24 cycle modulation signal. This control shall operate from a source impedance of 600 ohms ± 10%.

3.3.1.4.2 Connectors -

(1) Front Panel Connectors - The following front panel connectors shall be provided:

(a) Spin Frequency - A UG-657(*)/U BNC connector labeled "SPIN FREQ" shall be provided to allow measurement or use of the spin frequency reference. This signal is obtained from the output of the modulation control.

(b) System Trigger - A UG-492(*)/U BNC connector labeled "SYSTEM TRIGGER" shall be provided to connect the Radar Set system trigger to the 30 mc pulsed oscillator.

(c) Test Targets Trigger - A UG-492(*)/U BNC connector labeled "TEST TARGET TRIGGER" shall be provided to connect the 30 mc pulsed oscillator output to the Radar Set.

(2) Case Connectors - Two multi-pin connectors shall be provided on the side of the case to provide test connections to the aircraft test panel or the test bench harness. The connectors shall be identified as J101 and J102.

3.3.2 Cable Assembly, Special Purpose W-1 of AN/UPM-116 - The equipment shall be furnished with a 10-foot test cable to provide connections to the aircraft test panel. The connectors shall be identified as P101, P102, P2001, P2004, P2005, P2006, P2007 and P2009.

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3.3.3 Cable Assembly, Radio Frequency W-2 and W-3 of AN/UPM-116 - The equipment shall be furnished with two 10-foot coax cables to provide connections for the 30 mc pulsed oscillator and system trigger. The connectors shall be as follows:

Cable W2	P104, P2008	MS35168-88D
Cable W3	P105, P2003	MS35168-88D

3.3.4 Fuses - A 1.5 amp fuse, part number MS90077-12-1, and spare shall be included to protect the equipment in the event of electrical overload.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Preproduction Sample - Unless otherwise specified in the contract or order, a preproduction sample of the Test Set, Radar AN/UPM-116A shall be manufactured such that it is representative of the production lot in all respects. The sample will be tested as specified in section 4 herein and is for the purpose of determining that the contractor prior to his starting production is capable of yielding items that comply with the technical requirements of the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Tests Required - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Sample Tests

(3) Acceptance Test

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract.

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Preproduction tests shall include environmental tests in accordance with MIL-T-5422 to the extent specified herein, interference tests in accordance with MIL-I-6181 and drop tests for transit and combination cases in accordance with Specification MIL-T-21200.

4.2.3 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract.

4.3 Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and shall have successfully passed the individual tests.

4.3.1 Scope of Test - As a minimum, each equipment selected for sampling tests shall be subjected to the following test:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

4.3.2 Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Special Tests

(3) Life Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated at the conditions specified herein a total accrued running time of 6 hours. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a maximum double amplitude of 0.02 inches.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes as specified herein. The direction of vibration should be vertical to the normal mounting plane. The equipment shall be non-operating during the 10 minute vibration period. The 6-hour period of operation shall include that time necessary for perforı ing all equipment tests and calibrations. Only that time necessary to make a total accrued running time of 6 hours is required for operation after the 10-minute vibration period. During the 6-hour period of operation the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test continued. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.4.2 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.2.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require variations to assure satisfactory control.

4.4.2.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minute cycling in each of 3 planes)

(3) Humidity Tests

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4.4.3 Life Tests - When requested by the procuring activity, one unit shall be selected from current production equipment and shall have passed the Individual Test.

4.4.3.1 Scope of Test - A 200 hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.4.3.1.1. A performance check shall be made to determine the equipment is operating satisfactorily.

4.4.3.1.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal Room
Altitude	Normal Ground (0 - 5000 ft.)
Humidity	Room Ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	27.0 ± 2.0 volts

4.4.3.1.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3.1.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.4.3.2 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.4.3.3 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design of material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

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(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45.R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
NAVY DEPARTMENT

Chief, Bureau of Naval Weapons (RREN-5)

Navy Department

Washington 25, D.C.

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